CYBR125

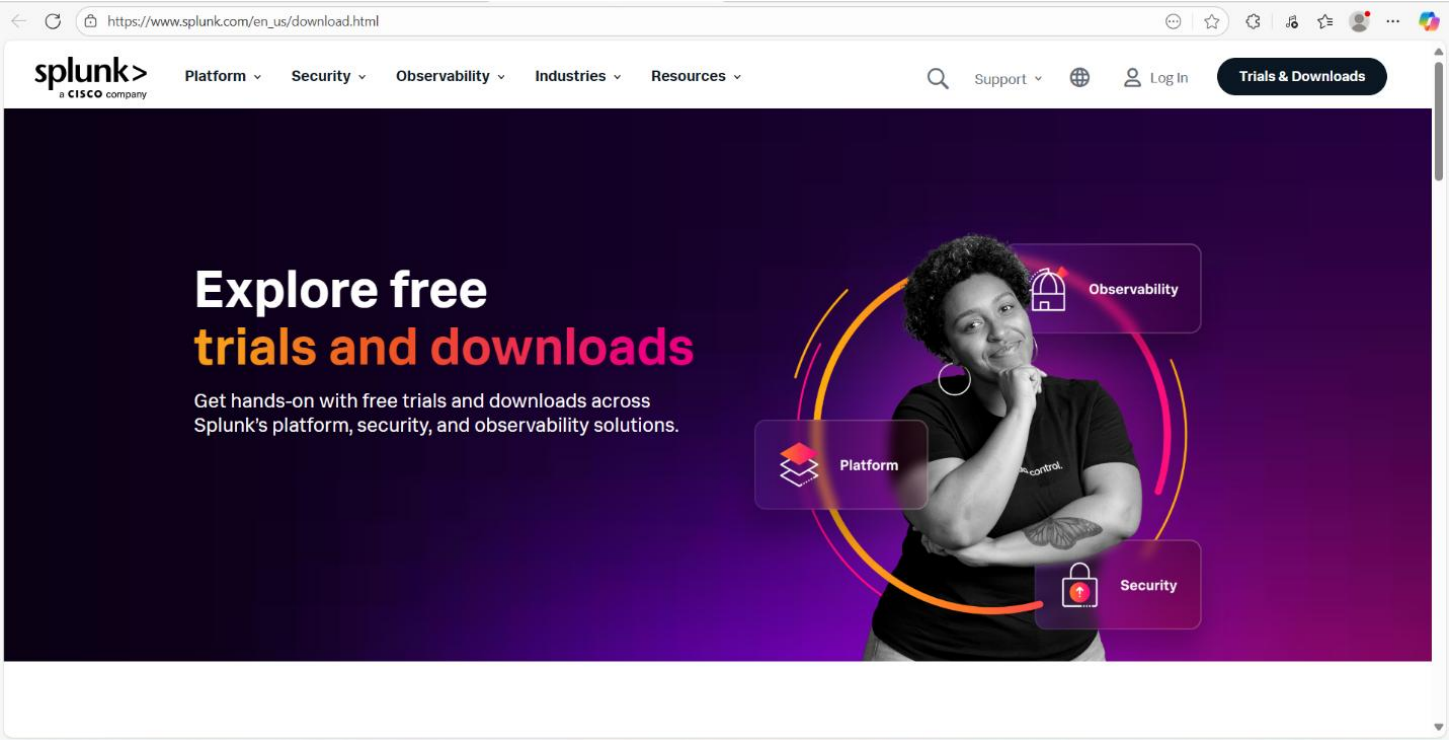
Security Devices and Appliances

LAB - 2

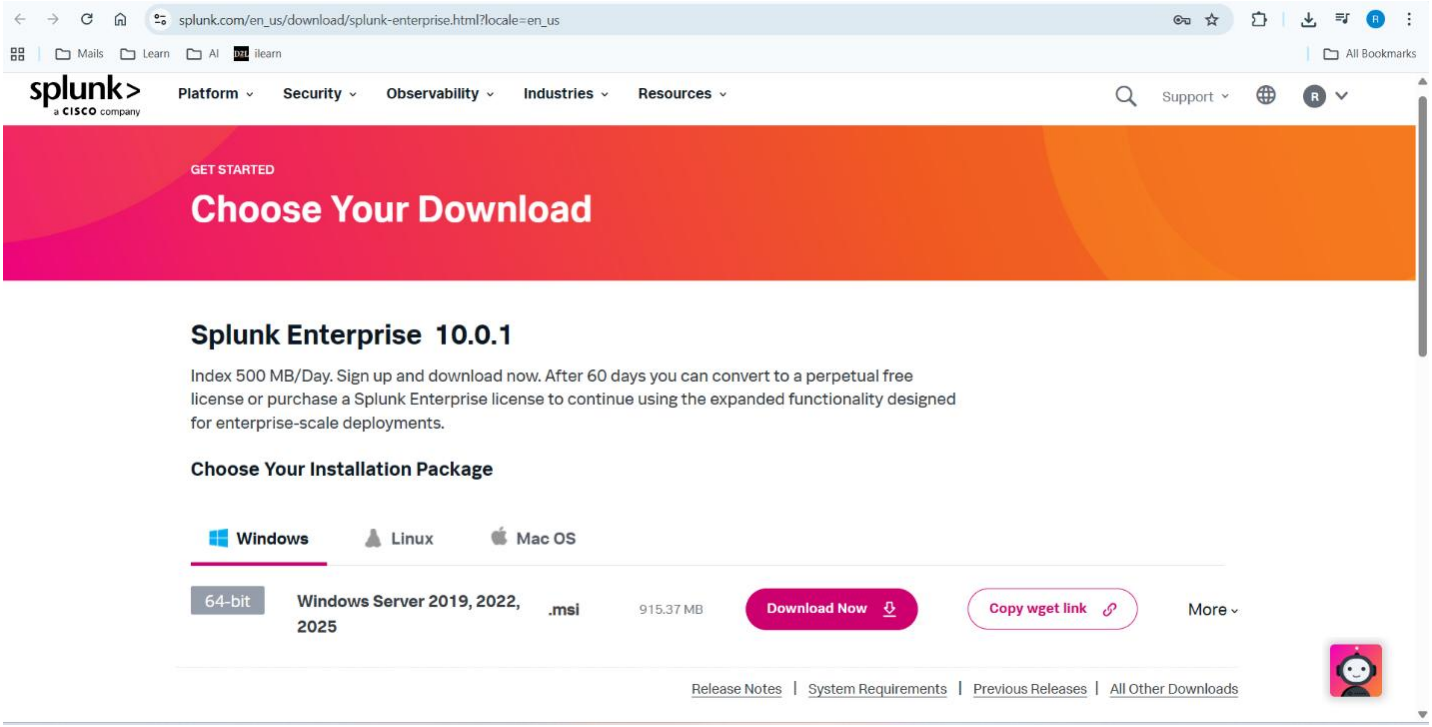
Ranjith kumar Murugan – A00191976

Anaz Lalu - A00201042

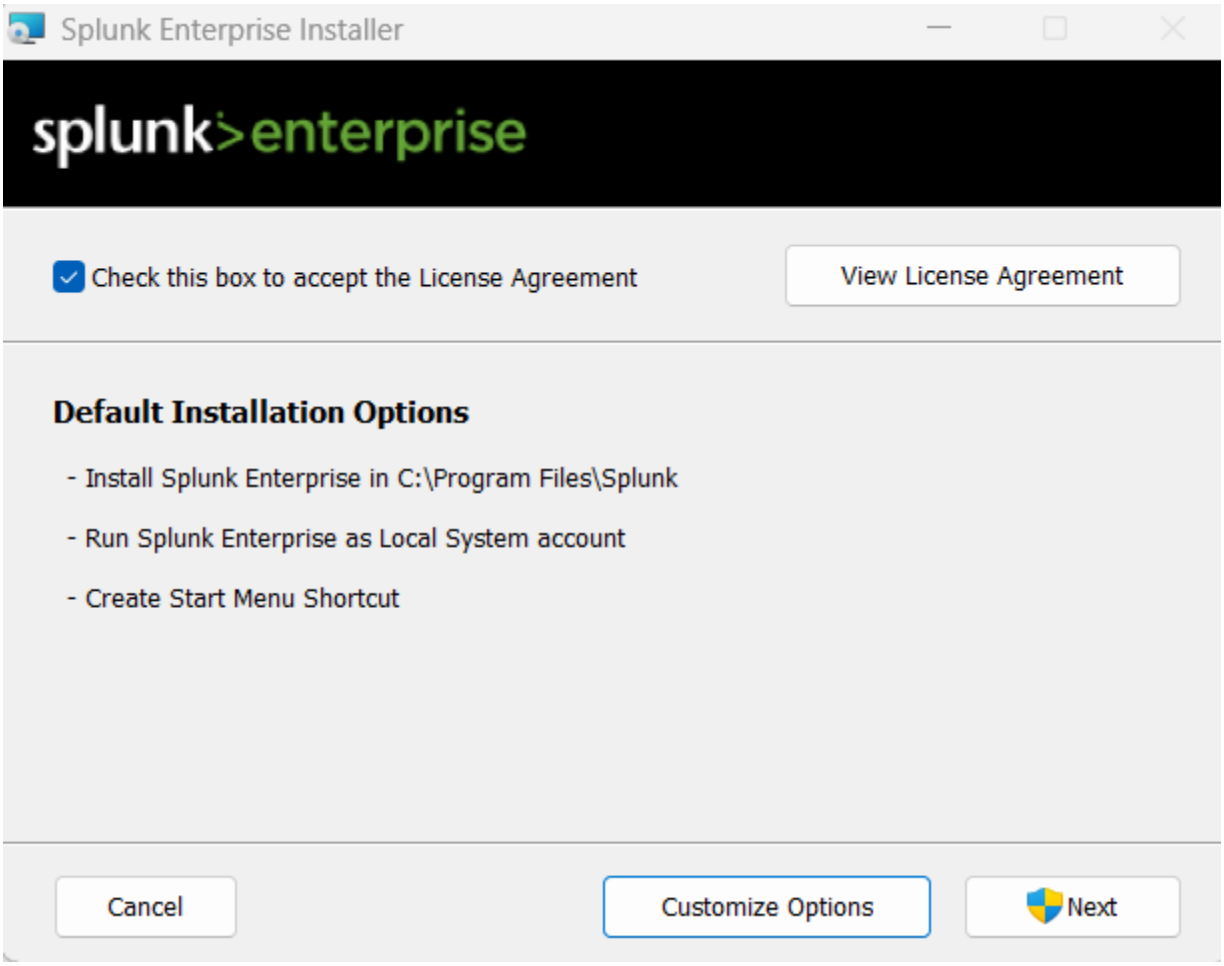
Installation process of Splunk



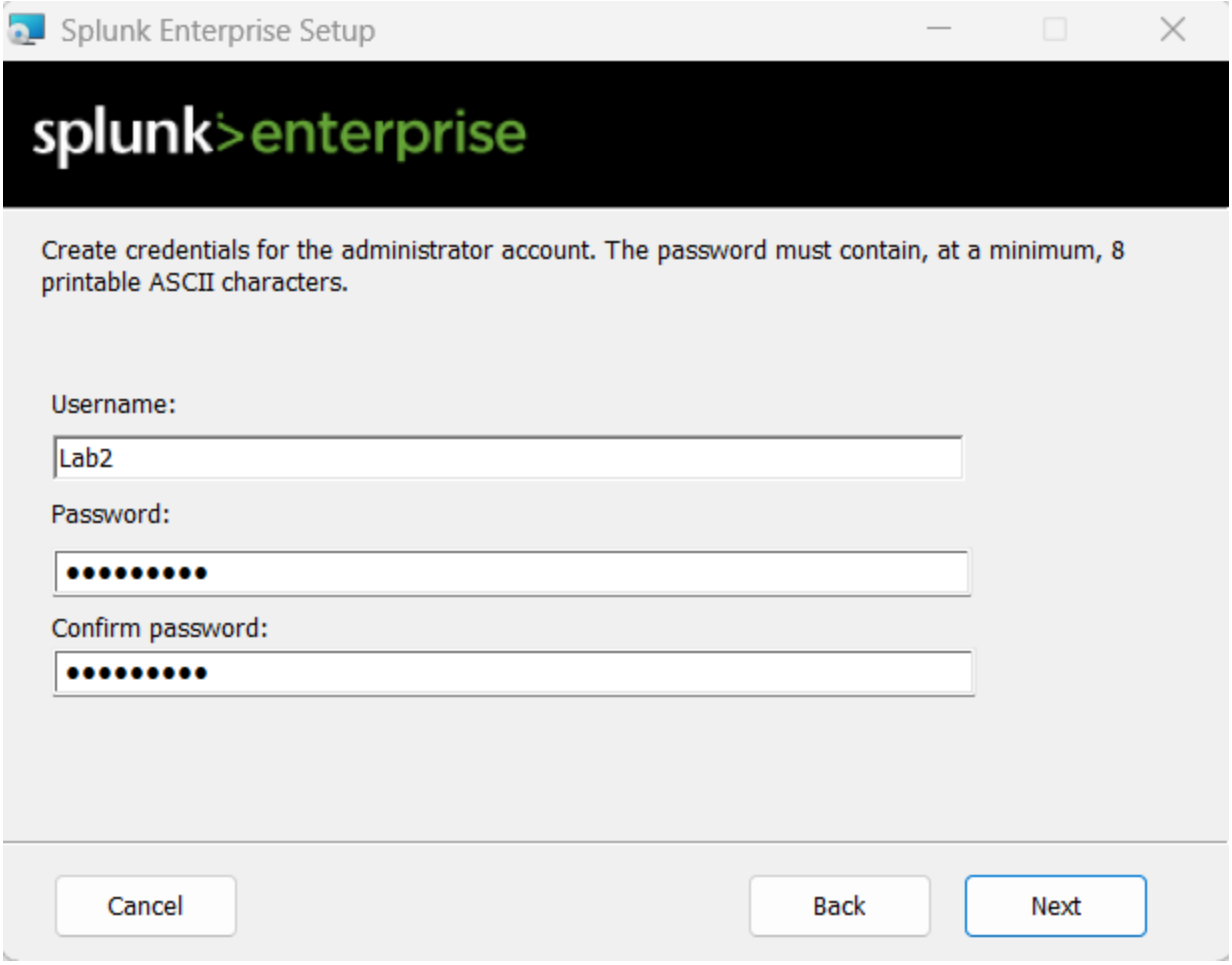
- Go to https://www.splunk.com/en_us/download.html, to download Splunk Enterprise software.
- Create an account and click on Trails & Downloads.



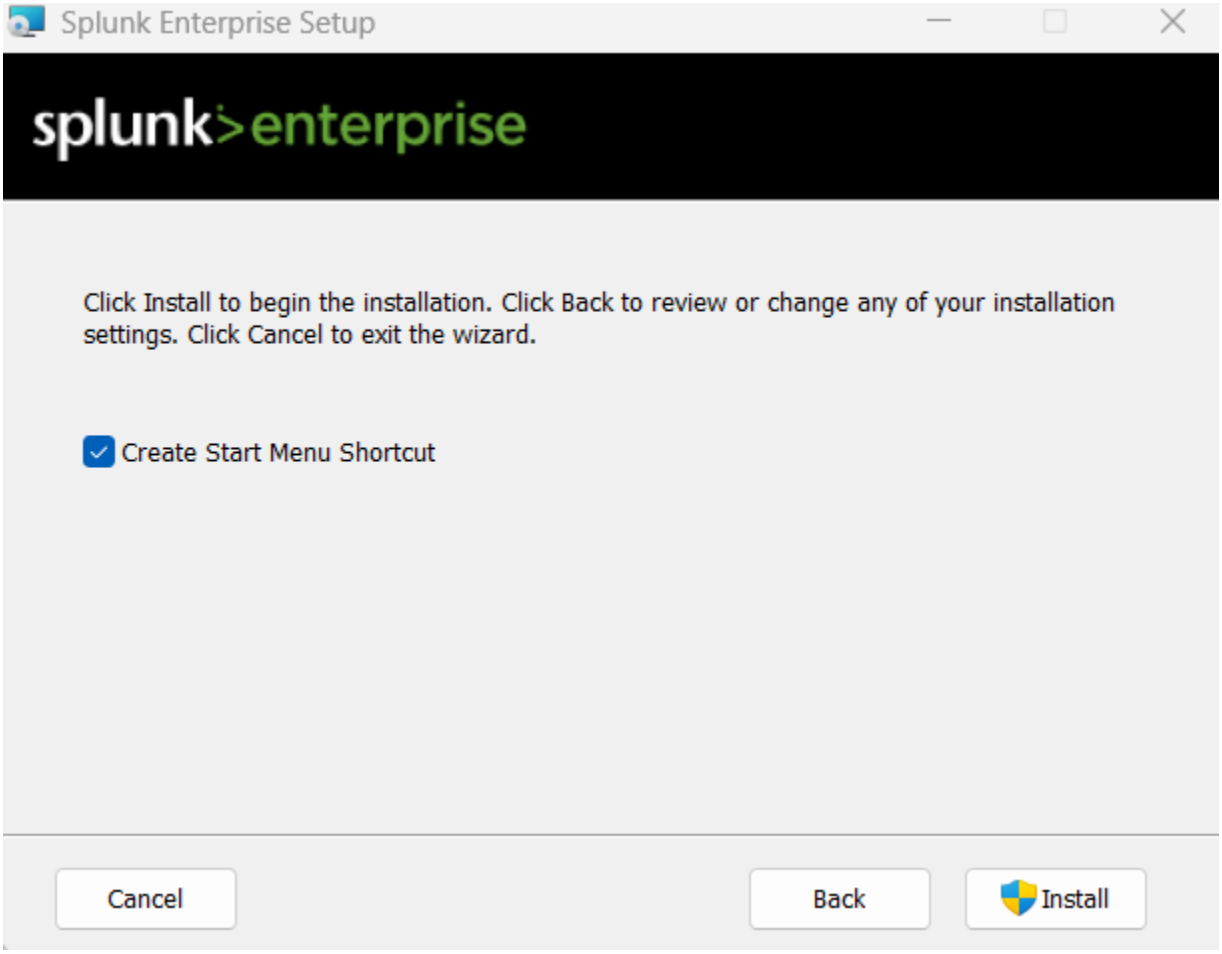
- Click on **Download Now** (this is for windows OS, choose yours accordingly). Once the file is downloaded, Open that to install.



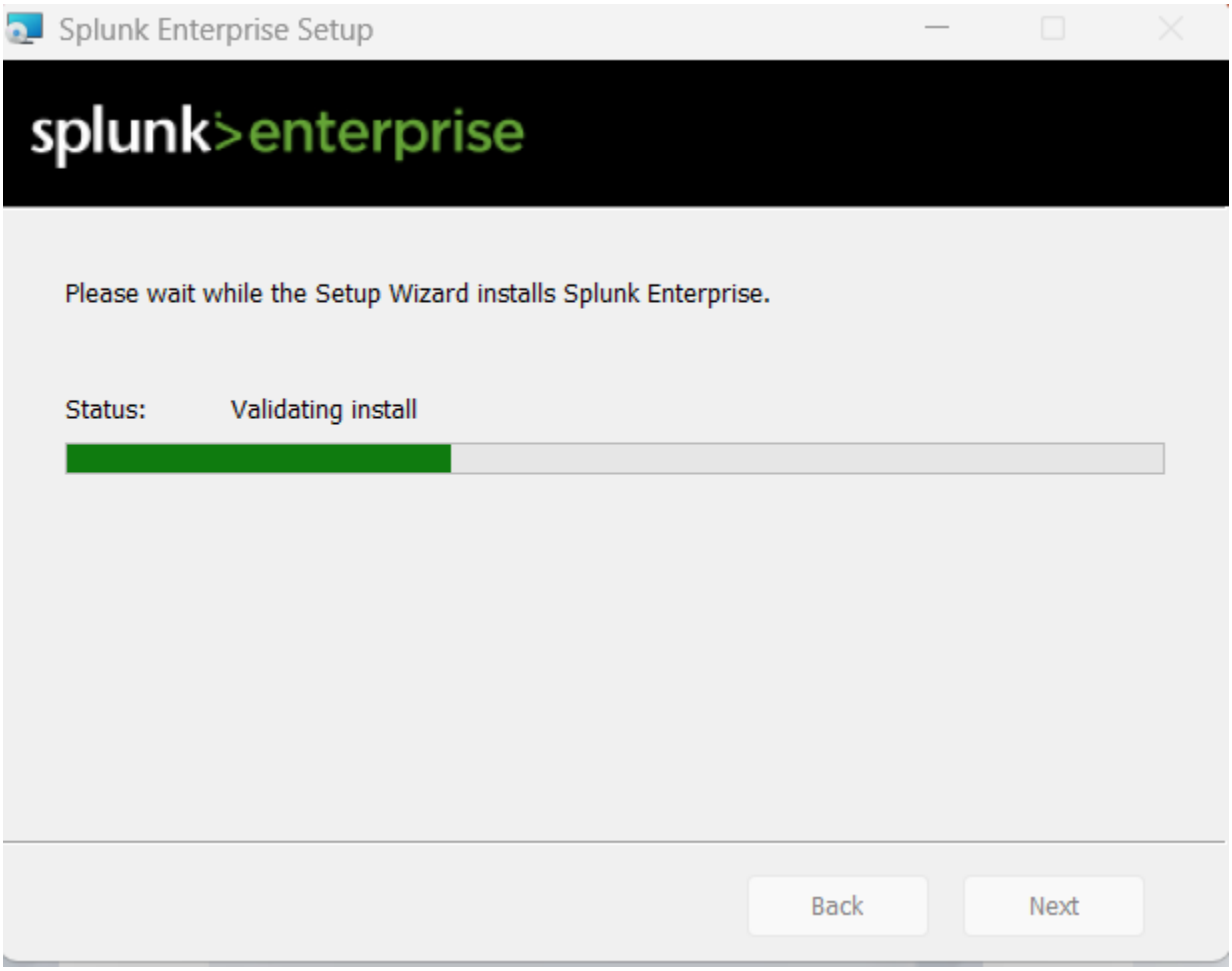
- Check the **Checkbox** to accept the License Agreement and click **Next**.



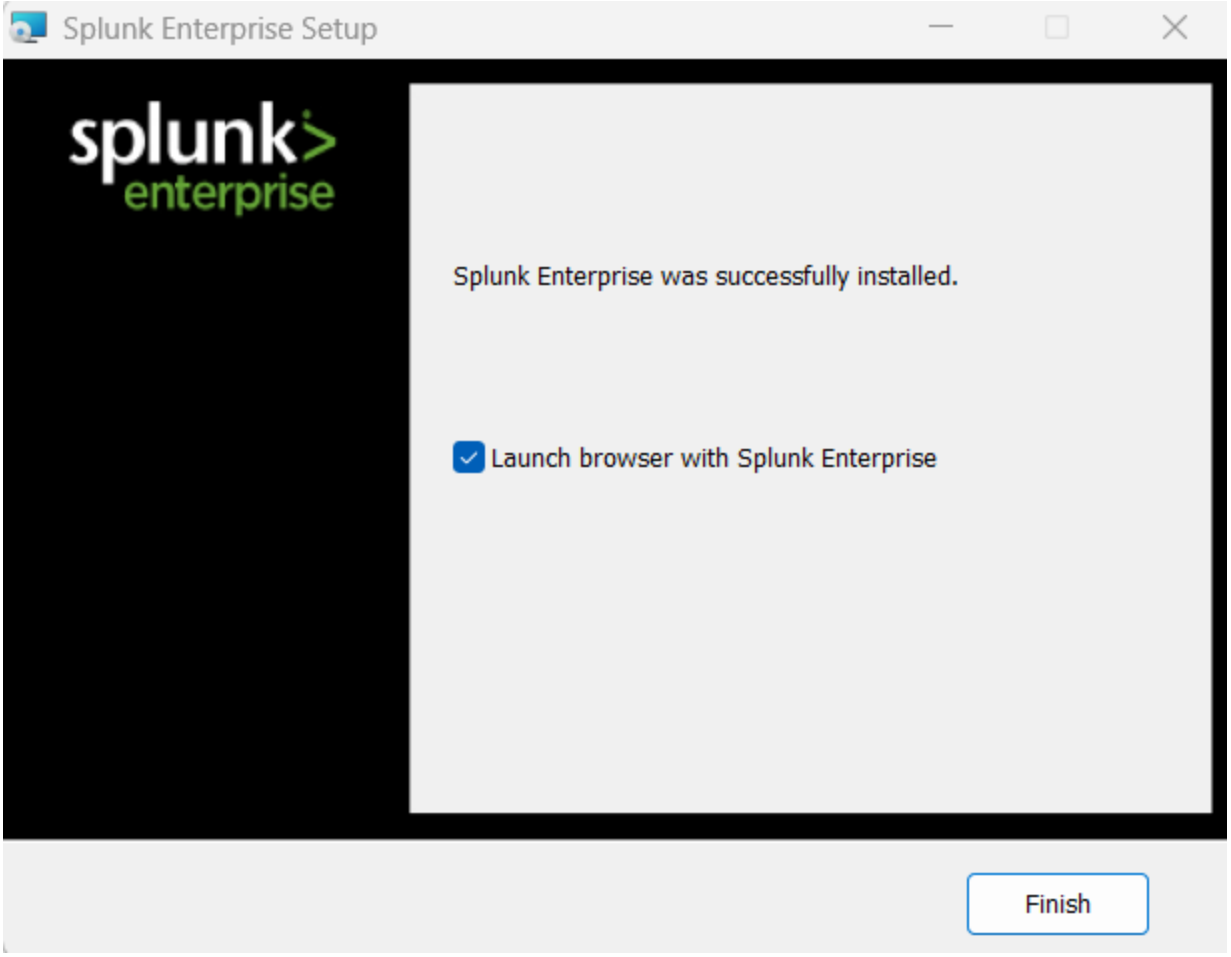
- Create a **Username** and **Password**, Click next.



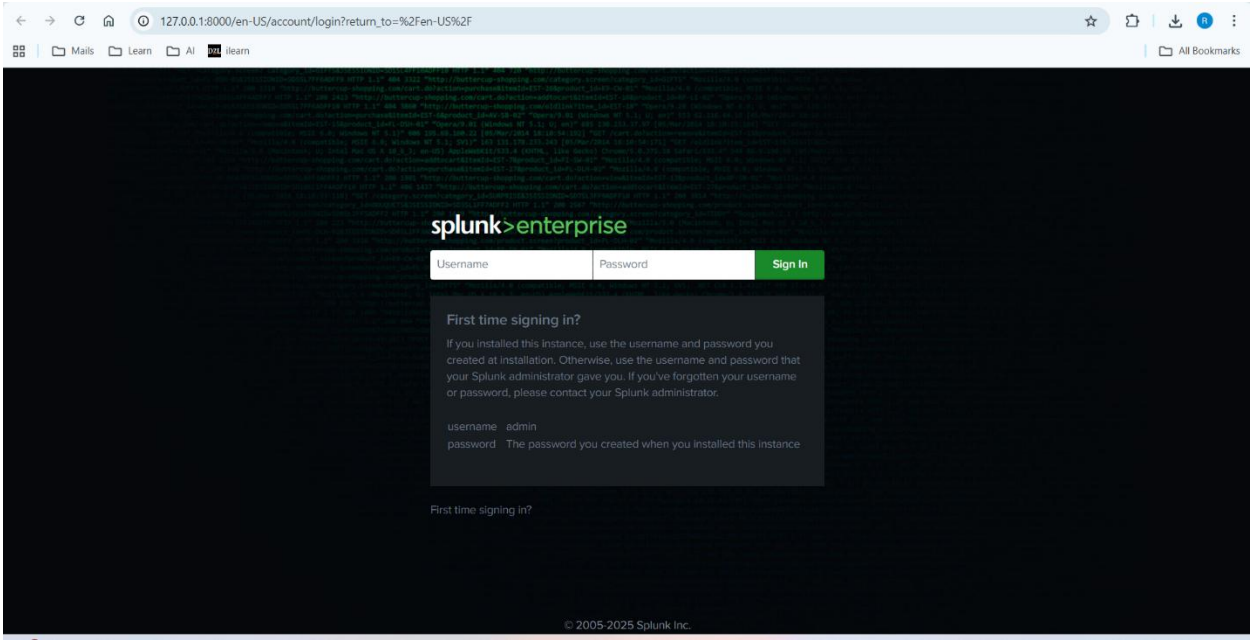
- Click **Install** option to start the process**.**



- The installation process will start. Wait until the process complete.

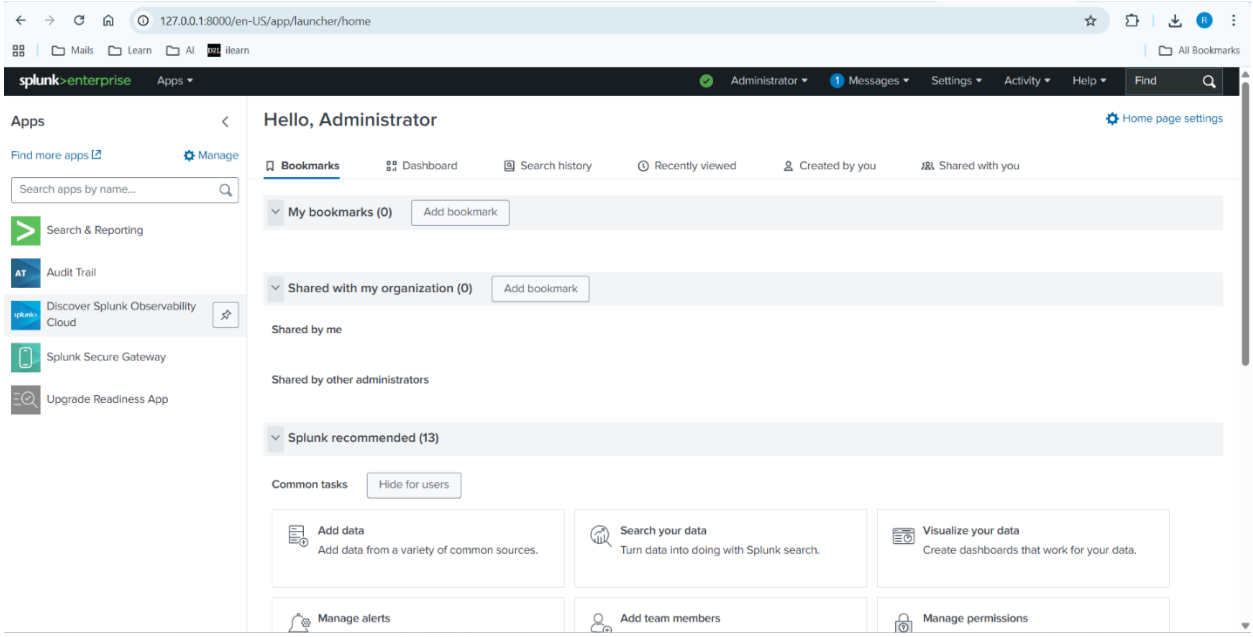


- Now we installed Splunk. Click **Finish**. It will direct us to the Splunk Enterprise Login Screen
- The below Screenshot is Login page.

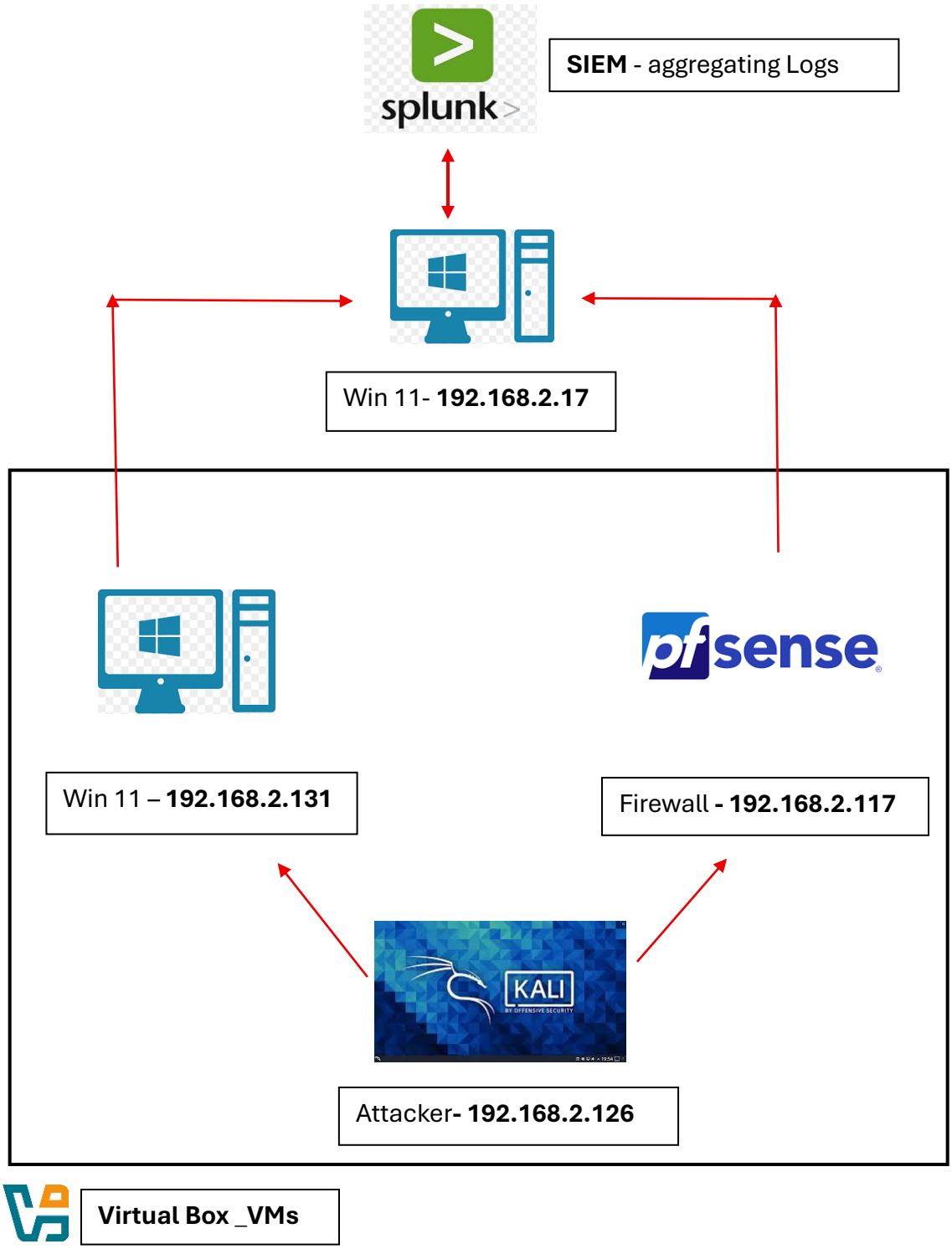


- Login using your credentials.

- The Below Screenshot is the **Post login screenshot** of Splunk.

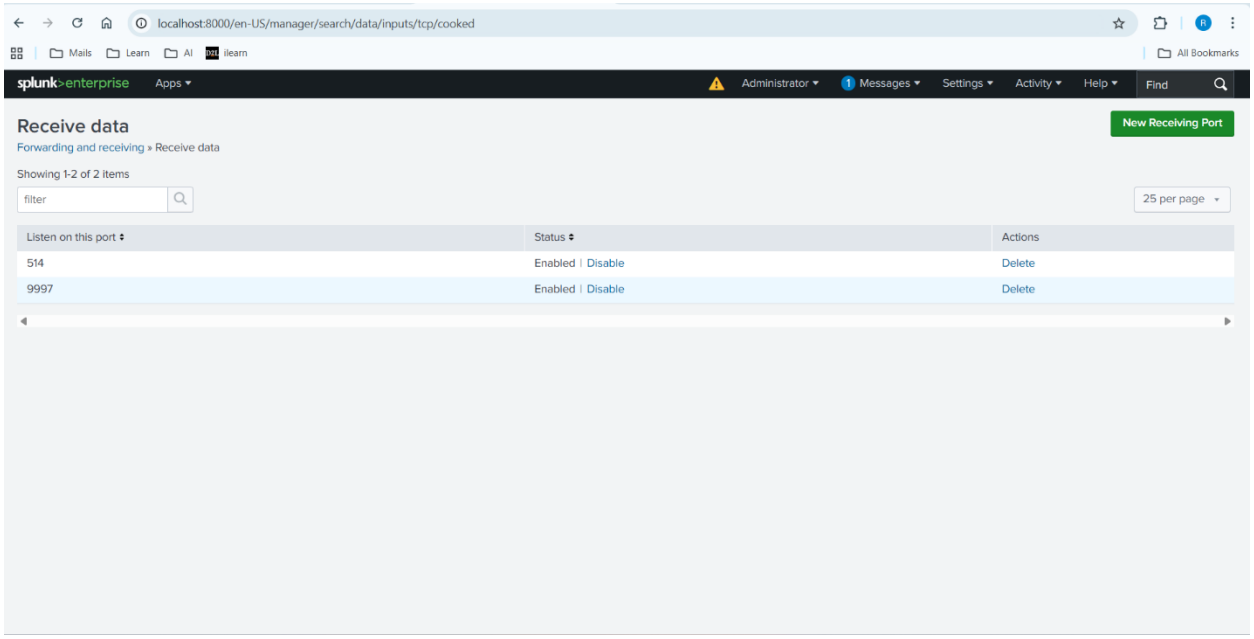


NETWORK DIAGRAM

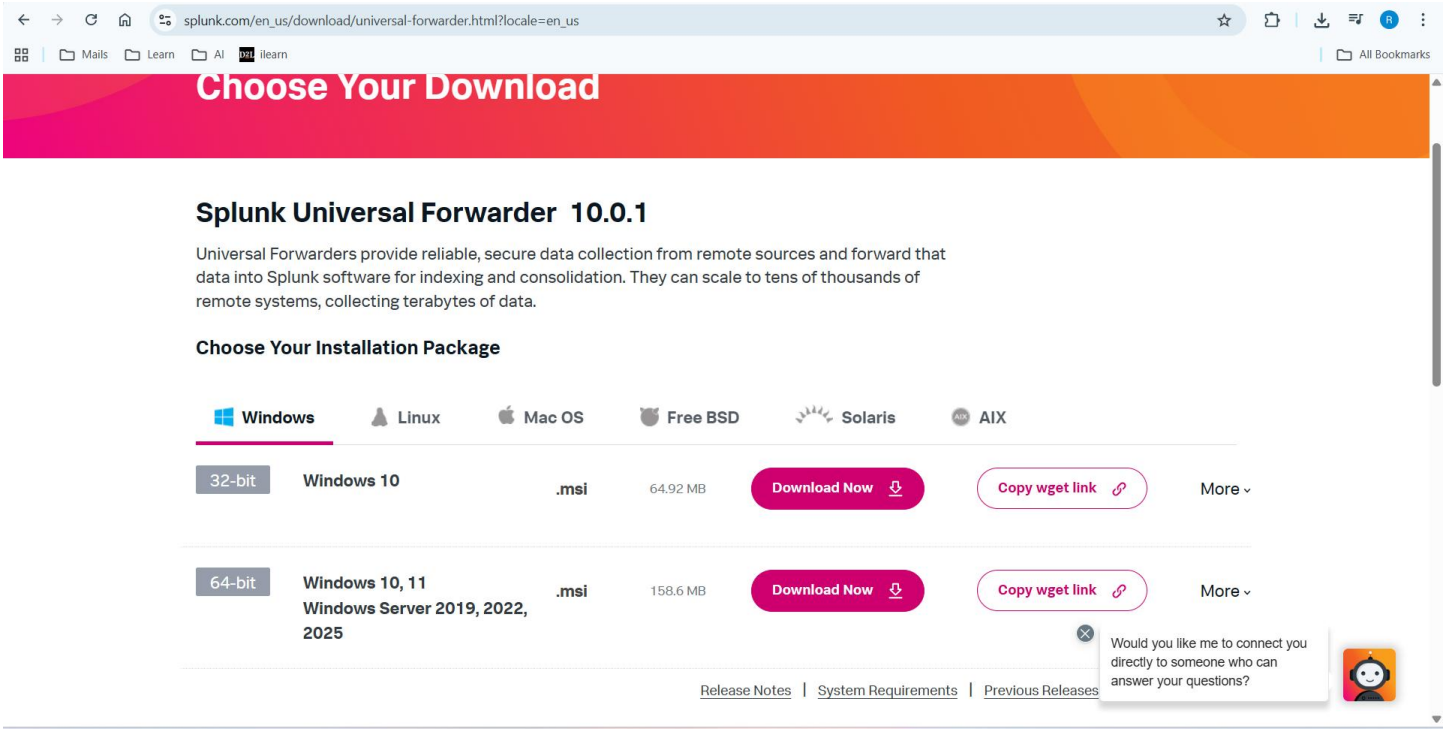


Attack Simulation and Detection:

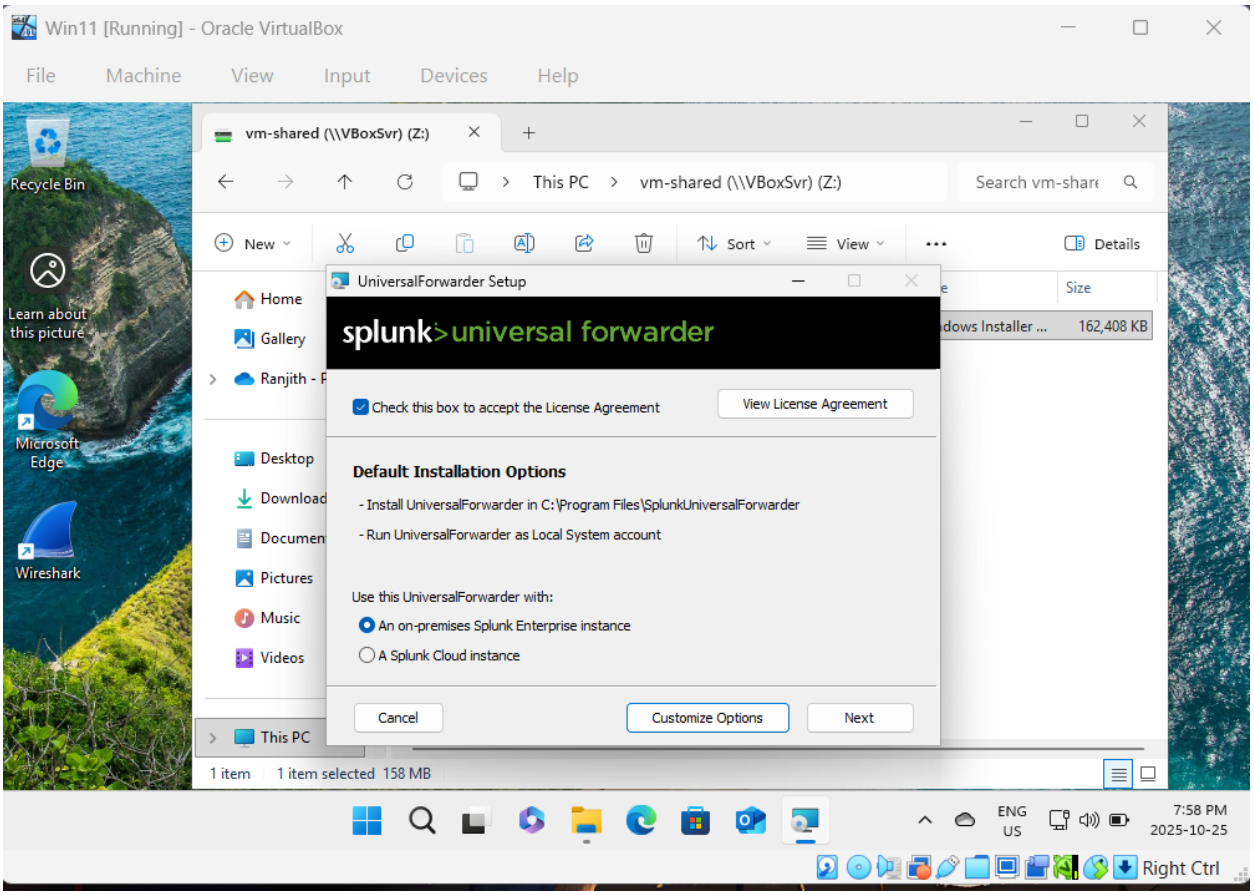
- Set the Listening port on the Splunk to receive data from Windows and Firewall.
- For example, I set 9997 for windows and 514 for Firewall.



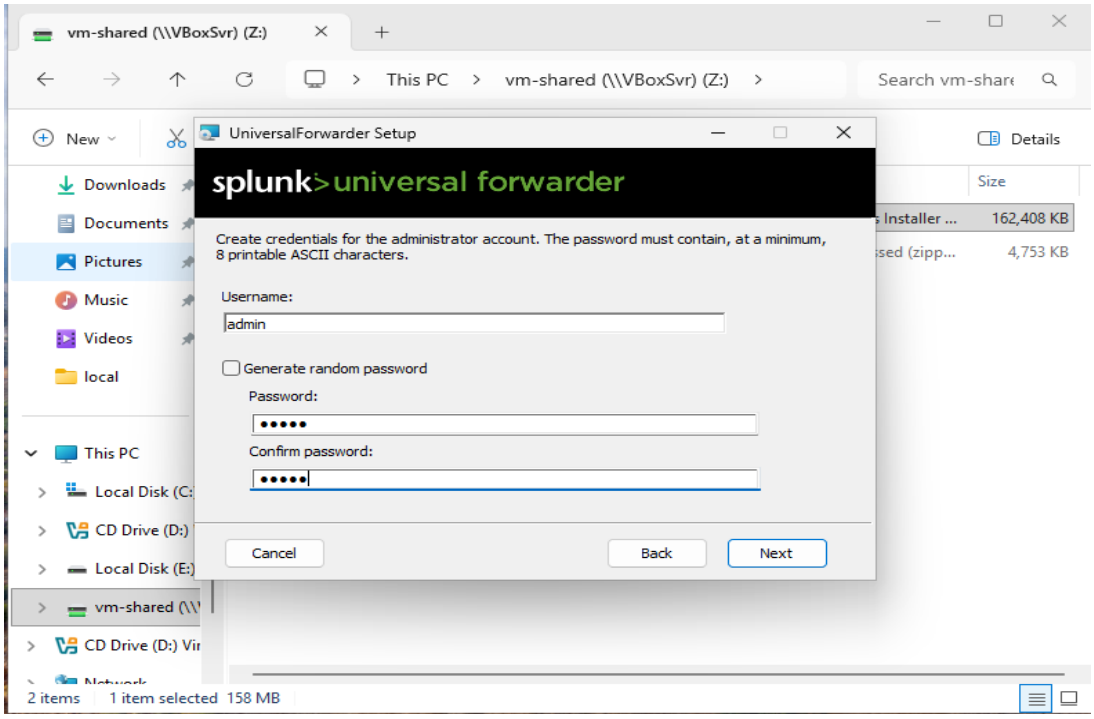
- Download Splunk Forwarder to forward logs from windows to splunk.
- I downloaded for windows 11
- Website -(https://www.splunk.com/en_us/download/universal-forwarder.html?locale=en_us)



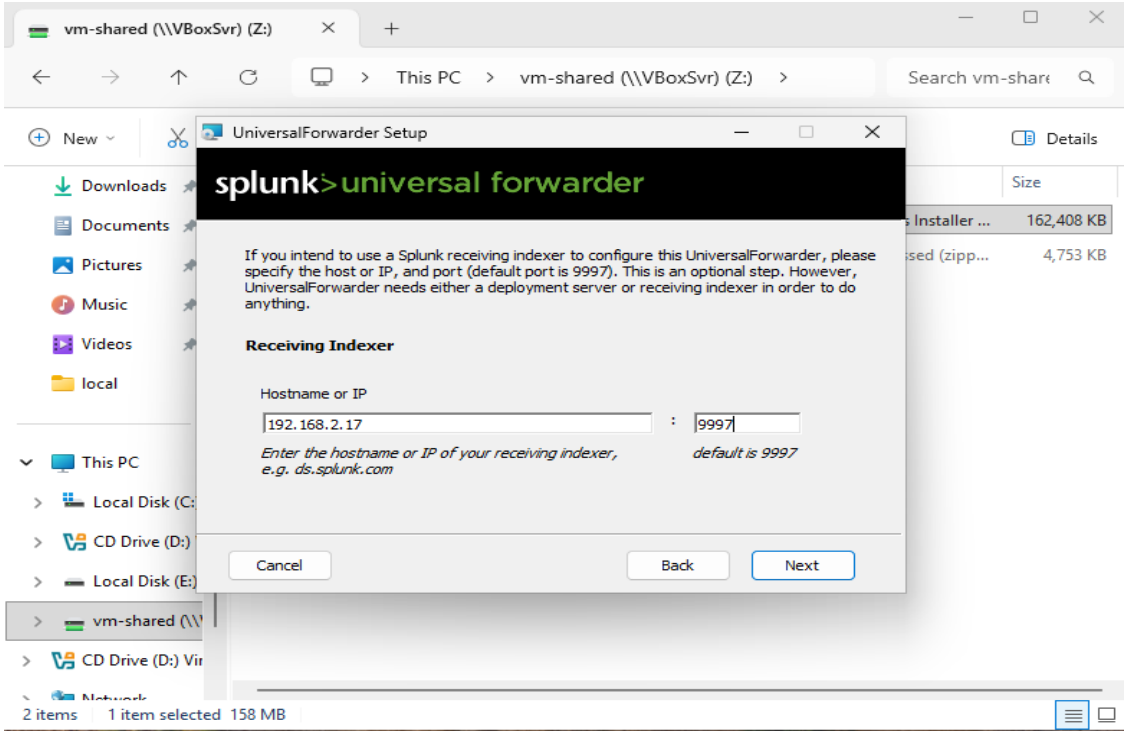
- Install the forwarder in Windows 11 to send logs to SIEM (Splunk).
- Check the Box to accept the agreement, then click next.



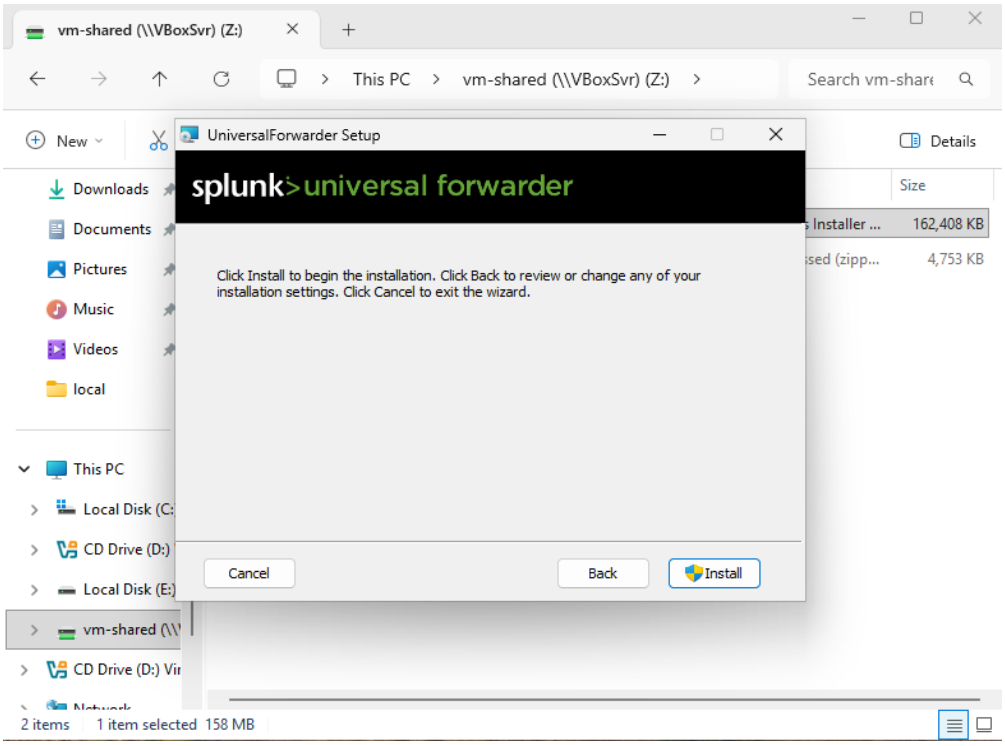
- Create a username and password for the forwarder.



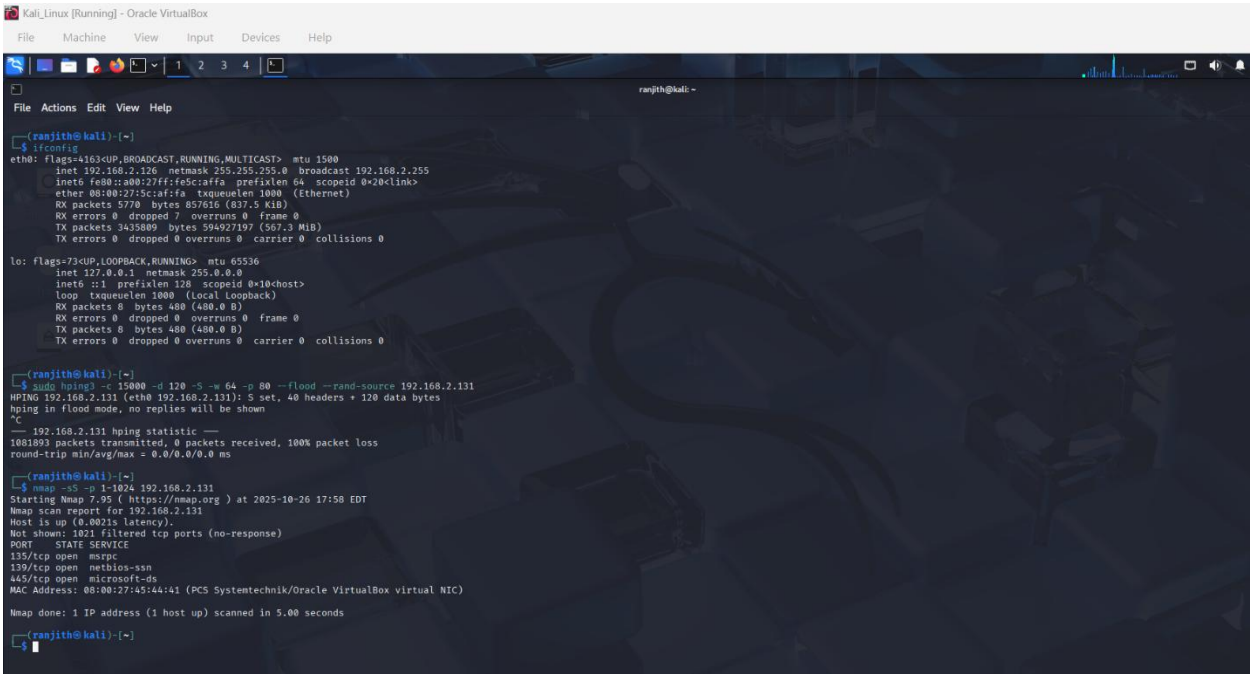
- You can skip the Deployment server window. Click Next.
- In the **Receiving Indexer,** give the **Splunk host IP** and **Listening port.** Click Next**.**



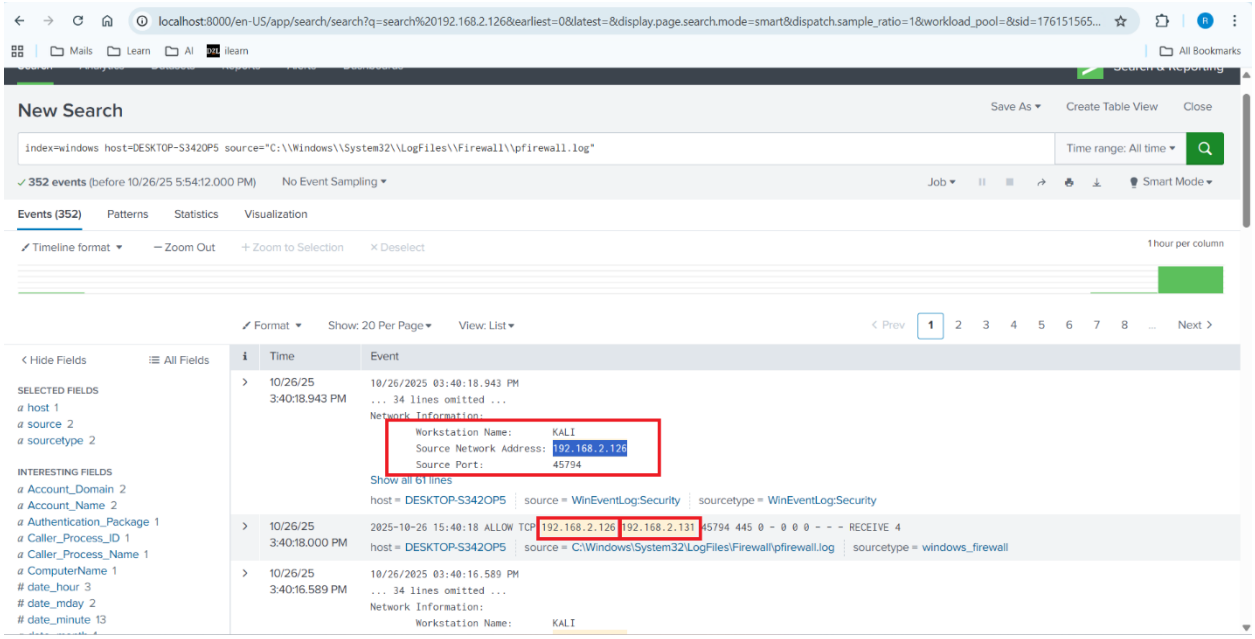
- Click Install, Then Finish.



Attack and Detection:



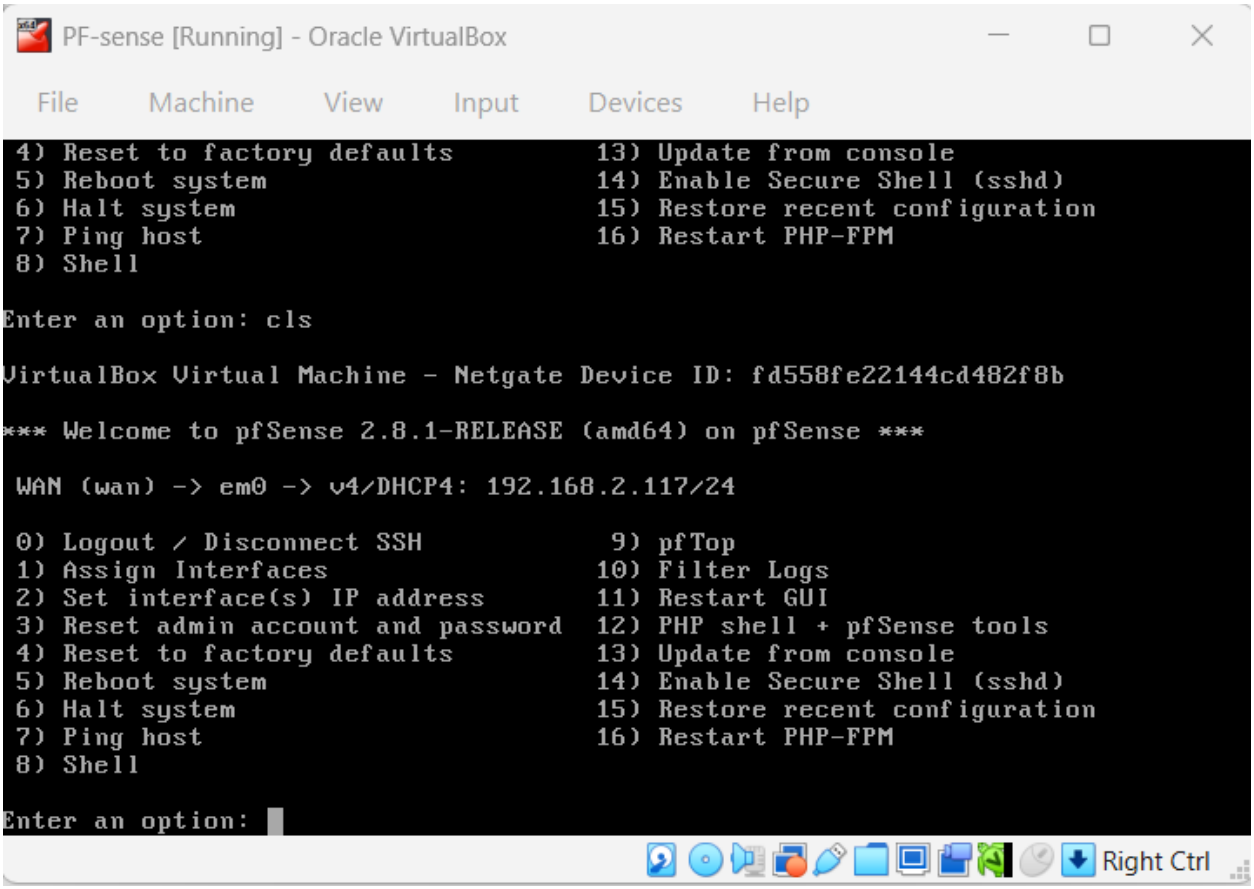
- I performed **hping3 Flood Attack** and **nmap Port Scan** against Windows 11.
- Kali IP- 192.168.2.126
- Windows IP – 192.168.2.131



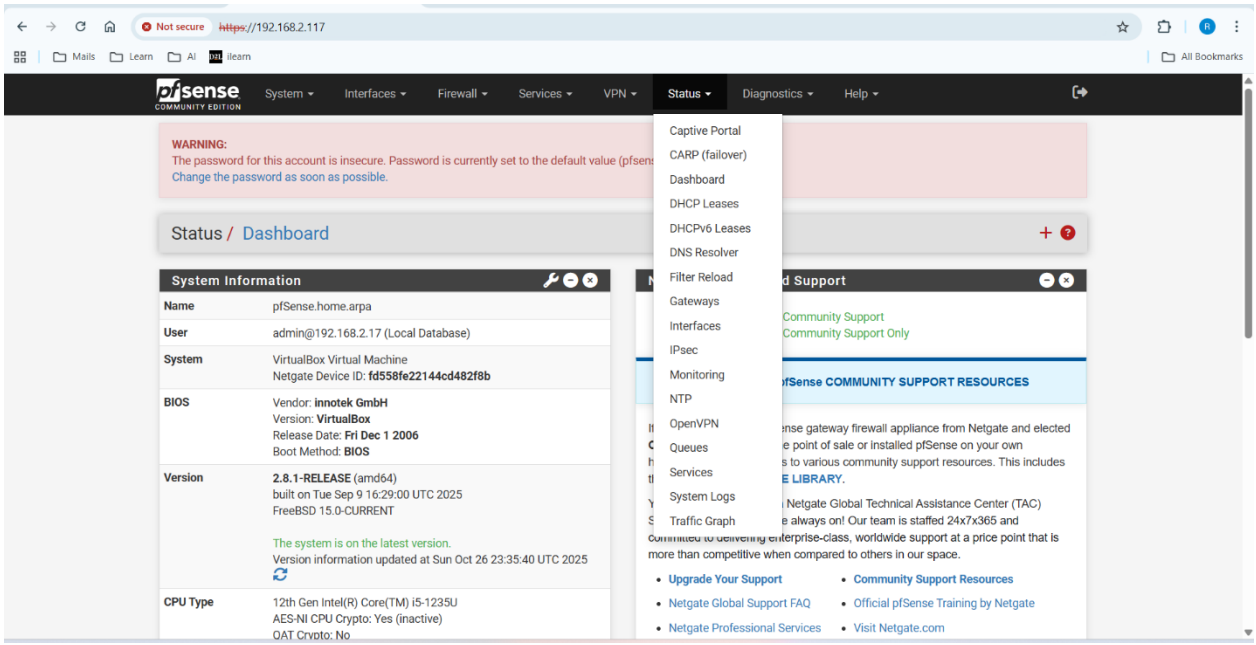
- **Search**- index=windows host=DESKTOP-S3420P5 source="C:\\Windows\\System32\\LogFiles\\Firewall\\pfirewall.log"

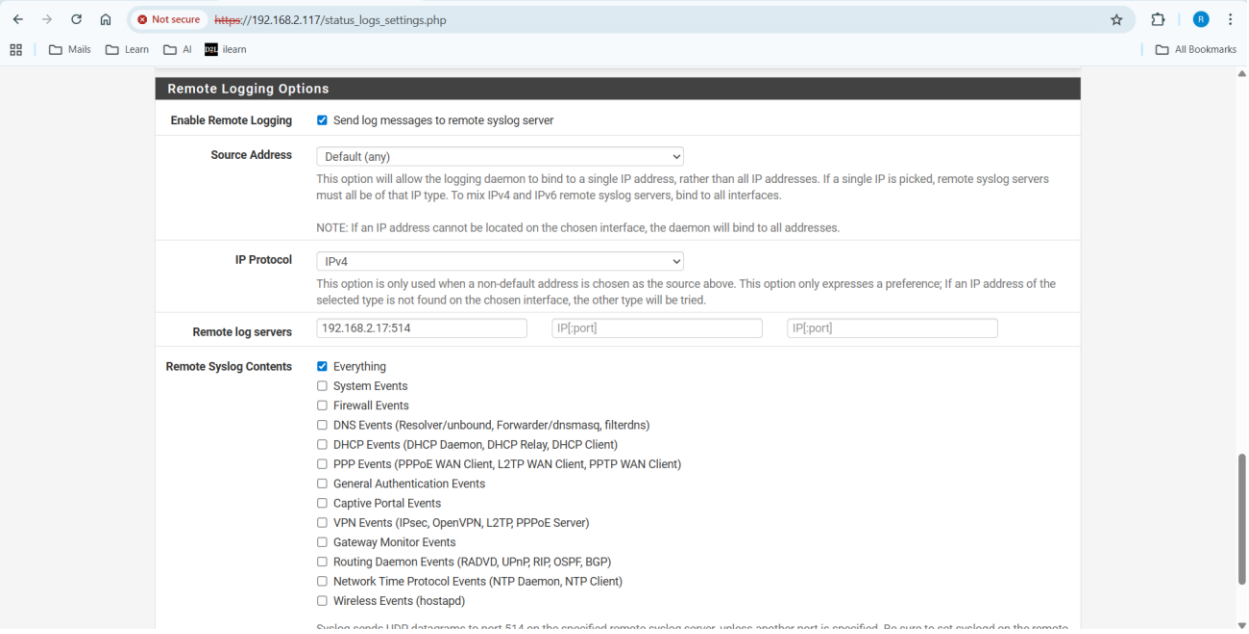
- When I check the **Windows Firewall log file** on Splunk **we can see the activity.**

PF Sense attack and Detection:

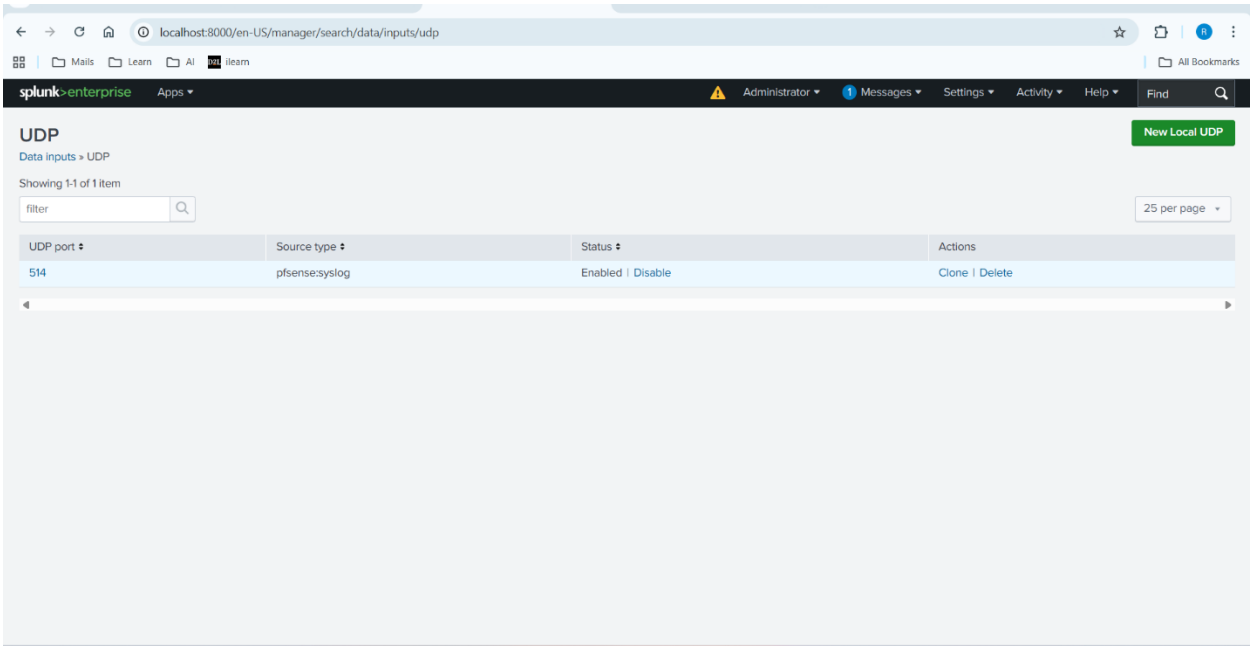


- PF sense IP – **192.168.2.117**
- Login to pfSense web **GUI → Status → System Logs → Settings → Remote Logging**

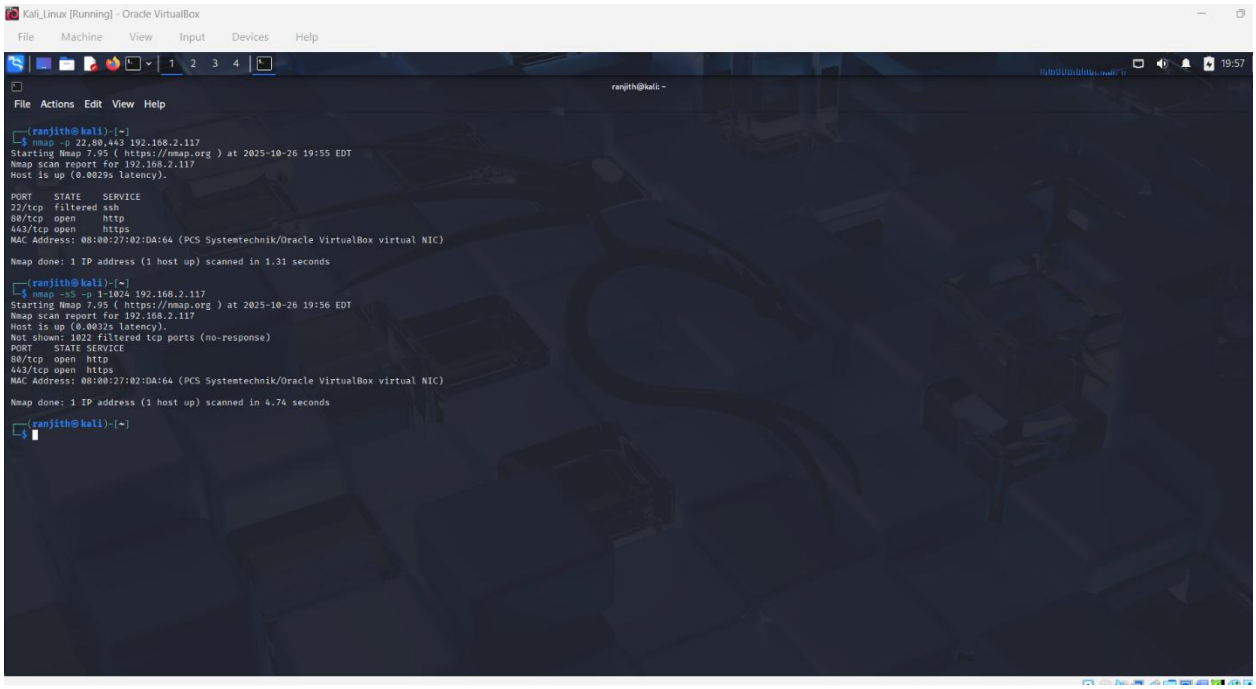




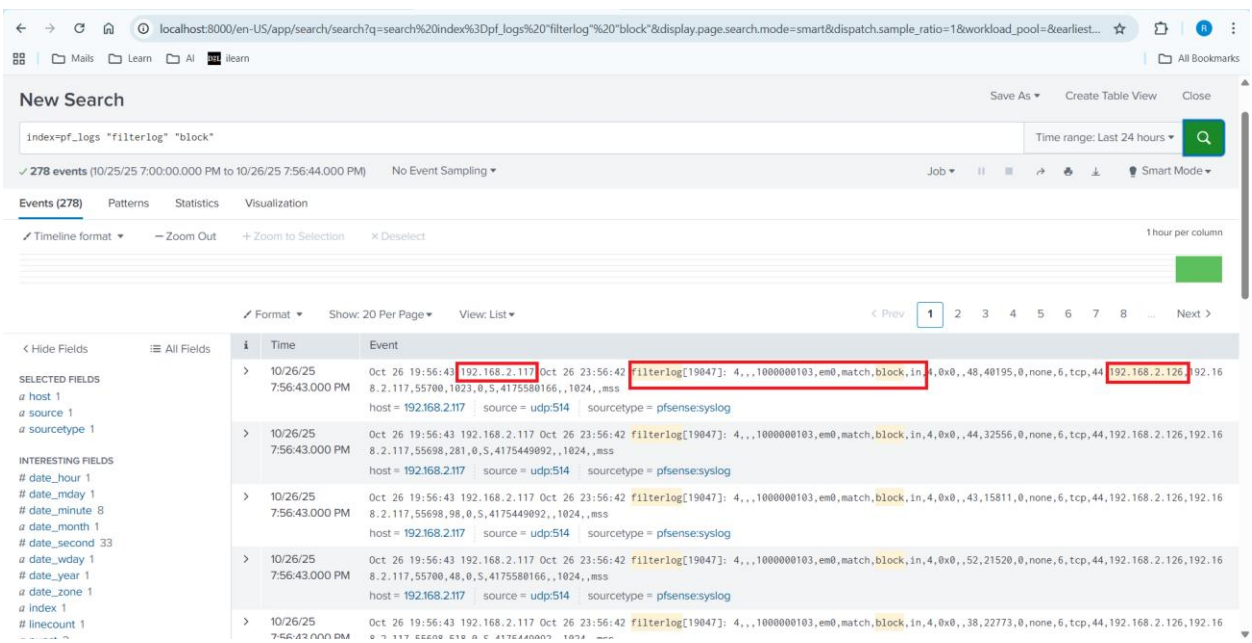
- Enable Remote Logging and add Remote log servers: (e.g. I added Splunk host IP and port **192.168.2.17:514**
- Choose your protocol (pfSense typically sends UDP), Then Check the boxes for which logs to forward (I select Everthing).



- On Splunk Side Go to Settings then Go to Data inputs create a UDP port to receive logs from PFsense.



- PF sense IP – 192.168.2.117
- I did **nmap -p 22,80,443 192.168.2.117**
  → a targeted port scan of ports 22, 80, 443. (Result: 22 filtered, 80 open, 443 open)
- And **nmap -sS -p 1-1024 192.168.2.117**
  → a SYN scan (half-open / stealth scan) of well-known ports 1–1024. (Result: 80 open, 443 open)



- Search – **index=pf_logs "filterlog" "block"**
- We can see pfSense successfully detected and blocked inbound traffic from your Kali VM. The logs appeared in the Splunk.
- **Kali IP**- 192.168.2.126